

**PROPOSAL FOR THE IMPACT OF IMMERSIVE TECHNOLOGIES
ON STEM EDUCATION IN SRI LANKA'S SECONDARY AND
TERTIARY CURRICULUM**

25-26J-418

Project Proposal Report

B.A.S.Hasith Bulathgama

IT22913760

B.Sc. (Hons) Degree in Information Technology Specializing in
Interactive Media

Department of Information Technology
Sri Lanka Institute of Information Technology, Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Signature
B.A.S. Hasith Bulathgama	

Signature of the Supervisor:

Date:

Signature of the Co-Supervisor:

Date:

ABSTRACT

The goal of this research mainly focuses on creating an Augmented Reality (AR) application for mobile phones targeting mathematics education for grade eight (8) curriculum of the Sri Lankan Education System. The idea is to get past those flat textbook pictures. Instead, bring in 3D stuff that's immersive and you can interact with. Built using Unity, the app gets optimized for cheap Android phones. It works offline too. That way, schools without much tech can still use it. Kids get to rotate shapes, slice them open, even unfold them. For algebra, they manipulate graphs on the fly. See how equations change the visuals right there. Abstract ideas become something you can touch and play with. Then there's this quasi-experimental study in actual classrooms. It checks if the app really works. Measures things like better scores, sharper problem-solving, more confidence, and kids staying engaged. Outcomes should mean deeper grasp of concepts. Stronger skills in seeing and spacing things out visually. More drive to learn math. Plus, it's the first real proof in Sri Lanka of how AR shakes up old-school teaching.

Keywords: Augmented Reality, Mathematics Education, STEM, Sri Lanka, Mobile Learning

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List of Abbreviations

Abbreviations	Description
2D	Two-Dimensional
3D	Three-Dimensional
AR	Augmented Reality
DDR	Design-Develop-Evaluate
GB	Gigabyte
R&D	Research and Development
RAM	Random Access Memory
SLIIT	Sri Lanka Institute of Information Technology
STEM	Science, Technology, Engineering, and Mathematics
VR	Virtual Reality

1 INTRODUCTION

The backbone of junior secondary math education in Sri Lanka comes down to state-issued textbooks. These books act as the main teaching tool, but they fall short when it comes to showing off the lively, multi-layered side of math concepts. Take geometry, for instance, with things like 3D shapes and transformations, or algebra involving graphing functions, and even data visuals. Complex stuff often gets boiled down to flat, two-dimensional drawings. That old-school method really gets in the way of building key skills, such as imagining space or thinking visually, which you need for real command of the topics. Rather than creating solid mental pictures of how these ideas work, kids end up just memorizing formulas and steps by heart. This leads to shallow understanding and trouble using what they've learned in fresh problem-solving situations.

On top of that, the shortage of properly equipped school labs makes the problem worse. It limits chances for hands-on exploration in math, a field that demands picturing abstract links and changes. Without interactive tools, getting a deep grasp of concepts becomes a real hurdle. Truth is, there's a big gap between the dry theory in textbooks and the practical know-how students actually need to build.

Around the world, Augmented Reality has stepped up as a game-changer in education. It's great at connecting abstract ideas to real experiences, as studies point out [9]. Recent reviews show how AR blends creative methods into learning, helping make tough math topics easier to approach and more fun [9]. Looking at meta-analyses, AR setups boost math performance in K-12 settings, with results depending on things like how long it's used, the subject focus, and the tech setup [10]. Basically, AR layers interactive 3D models over everyday learning items, turning hidden processes into something you can see and tweak on the fly. It opens up intuitive ways for students to dive in and explore, and research notes that its impact in math teaching hinges on factors like kids' ages, how it fits the curriculum, and the exact math area covered [10].

This report lays out a detailed, research-backed R&D strategy for a homegrown AR app tailored to tackle these teaching and tech issues in Sri Lanka. Pulling from fresh studies on weaving AR into creative math lessons [9], the plan seeks to shift learning from rote repetition to vivid visualization and true conceptual depth.

1.1 Background and Literature Survey

< Background or Problem Domain Literature Survey >

In Sri Lankan schools, a bunch of places are short on lab stuff and hands-on things for math class. You know, math needs kids to get those moving ideas, like graphs that shift around, shapes that change, and data that gets complicated. When there's no gear, students just picture it all in their heads. That way often messes with really understanding the basics.

AR, or augmented reality, pops up as a fix that's used all over the world for this kind of problem. It links those abstract math bits to real life by slapping digital layers on everyday tools. Basically, it puts 3D models that you can mess with right over normal objects, making invisible stuff visible and touchable. Say you have a simple cone drawing. AR turns it into a 3D thing kids can spin, cut, or stretch right there (Fauzan et al., 2025) [7]. Feels way more real and fun than just staring at book pictures, especially in geometry where seeing space matters a lot.

Reviews from all over back up how AR helps in education. One big look at studies from the last twenty years on AR and VR shows it boosts learning, with better scores, more interest, sharper focus, and stronger memory (Bulut & Ferri, 2023) [1]. Newer roundups of papers say the same, proving AR ramps up math teaching by getting students more involved and responsive (Ivan & Maat, 2024) [6]. Those perks hit hardest in math and science, where picturing things clear is key. AR cuts the brain work of turning 2D sketches into 3D ideas, so kids jump straight into the concepts and build better space sense.

Studies show AR works in different math areas. Adding it to lessons improves results a lot if done right (Sirakaya & Sirakaya, 2023) [5]. Like in Germany's cleARmaths project (Schutera et al., 2021) [3], or apps focused on geometry that get kids interacting more (Fauzan et al., 2025) [7]. Plus, these tools spark creative math thinking and new ways to approach problems (Kugelmeyer & Elschenbroich, 2023) [8]. All of it shows how AR simplifies hard topics with real, hands-on versions of abstract stuff.

On top of that, software like Unity gets popular because it runs on all kinds of devices. Marker tracking ties printed images to virtual extras pretty well. But hey, tech alone doesn't cut it. Folks talk about the last-mile issue in edtech, where plans flop even with good ideas (Chonchaiya & Srithammee, 2025) [2]. For AR to stick in class, it has to work on what students already have, slot into regular lessons, and teachers need real training. Papers stress that pulling off AR means thinking hard about tech setup and how it fits teaching (Ivan & Maat, 2024) [6]. Skip those practical bits and support, and it usually fails.

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Background or Problem Domain
Literature Survey

Turns out AR helps the kids who struggle most. Top students often already picture things sharp in their minds (Elsayed & Al-Najrani, 2021) [4]. For ones stuck rote-learning, AR gives concrete ways in, like touching the ideas. Think of a flat net folding to a cube. AR walks through it, fills mental gaps, and evens out differences between kids (Chonchaiya & Srithammee, 2025) [2]. Research backs that AR boosts geometry grasp through interactive visuals (Fauzan et al., 2025) [7]. It also lifts creative math skills big time (Kugelmeyer & Elschenbroich, 2023) [8], and reviews show steady wins in all sorts of school settings (Ivan & Maat, 2024) [6].

Our This project pulls from all this, covering tech side and how people learn. The pile of studies, from big reviews (Bulut & Ferri, 2023 [1]; Ivan & Maat, 2024 [6]; Kugelmeyer & Elschenbroich, 2023 [8]) to hands-on tests (Sirakaya & Sirakaya, 2023 [5]; Elsayed & Al-Najrani, 2021 [4]; Fauzan et al., 2025 [7]) and real-world rollout info (Schutera et al., 2021 [3]; Chonchaiya & Srithammee, 2025 [2]), sets up a strong base for making an AR tool that fits Sri Lankan schools just right.

1.2 Research Gap

International research has already proven the benefits of using augmented reality (AR) in education. However, in Sri Lanka, we don't know if these benefits will apply, and there are some specific local challenges that need to be solved first. This project is designed to address those challenges and create an AR tool that works for Sri Lankan classrooms.

- **Curricular Alignment Gap:** Studies in other countries, like Germany and Saudi Arabia, have shown that AR can really help students understand math better. But these projects are often too generic and don't align with a specific country's syllabus. This is a problem because teachers need tools that fit right into their existing lesson plans.

Our project will fix this by directly linking AR models to the official Grade 8 Sri Lankan mathematics syllabus. For instance, when a student points their phone at a diagram of a cylinder or a bar chart in their government-issued textbook, an interactive 3D model will pop up. This makes sure that the AR isn't just a cool distraction; it's a practical and relevant part of the curriculum.

- **Technological Gap (Low-Resource Contexts):** Most AR research assumes that students have access to super fast internet and powerful, expensive devices. That's just not the reality for many students in Sri Lanka. They often use low-end Android phones with limited memory, and internet can be unreliable.

To make AR truly accessible, our app will be lightweight, offline-ready, and designed to work on modest devices. This means that students in rural or semi-urban schools won't be left out just because they don't have the latest technology. We're creating a solution that works for everyone, not just a privileged few.

- **Empirical Validation Gap (Sri Lankan Context):** We don't know for sure how AR will be received in Sri Lankan classrooms. Our education system is very focused on exams, which often encourages students to memorize formulas instead of exploring and understanding concepts. Plus, many schools don't have labs or interactive resources, so students rarely get to see math in a hands-on way.

This project will fill this gap by conducting the first quasi-experimental study of its kind in Sri Lanka. We'll compare how a group of Grade 8 students learns with traditional methods versus a group using our new AR tool. The results will provide solid evidence and a unique dataset from a South Asian perspective, which can contribute to the global research on AR in education.

- **User-Friendliness and Accessibility Gap:** A lot of the AR apps out there are too complicated, too expensive, or just don't work smoothly. This makes it tough for teachers to use them and for students to stay engaged. We've heard these frustrations from educators.

Our app is being built from the ground up to be simple, low-cost, and user-friendly. It will work with the devices students already have and integrate seamlessly with their textbooks. The goal is to create a tool that is so practical and easy to use that both students and teachers will actually want to use it.

<some description to guide the reader to the table/figure in the next page.>

AR Products	Textbook Integration	Offline Functionality	Low-End Device Support	Sri Lankan Curriculum	Grade 8 Specific	Free Access
GeoGebra AR	✗	✗	✗	✗	✗	✓
MERGE EDU	✗	✗	✗	✗	✗	✗
zSpace	✗	✗	✗	✗	✗	✗
Labster	✗	✗	✗	✗	✗	✗
cleARmaths	✓	✗	✗	✗	✗	✓
Proposed System	✓	✓	✓	✓	✓	✓

Table 1: System Gap

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Research Study	Identified Research Gap	My Research Contribution (Grade 8 Mathematics - Sri Lanka)
Chonchaiya & Srithamsee (2025) <i>AR for Geometry Learning (Thailand - Vocational School)</i>	Focused on vocational students aged 18-35; limited to basic geometric concepts; no curriculum alignment with national secondary education standards	Targets Grade 8 secondary students (13-14 years) using official Sri Lankan curriculum alignment with government-issued textbook integration
Bulut & Borromeo Ferri (2023) <i>Systematic Literature Review on AR in Mathematics Education</i>	Identified lack of curriculum-specific AR tools; most studies use generic AR applications not tied to national textbooks or syllabi	Creates direct textbook-to-AR mapping using Sri Lankan Grade 8 mathematics textbook diagrams as AR markers for curriculum-aligned 3D models
Schutera et al. (2021) <i>AR for Middle School Mathematics (Germany) - Vector Geometry for Grade 12</i>	Focused on advanced vector geometry for Grade 12; requires Android devices with higher specifications; limited to German educational context	Targets foundational Grade 8 concepts (basic geometry, algebra, data representation) with preloaded assets optimized for 2-3GB RAM devices, addressing Sri Lankan infrastructure limitations
Elsayed & Al-Najrani (2021) <i>AR for Middle School Mathematics (Saudi Arabia)</i>	Limited to specific chapter (Area & Volume); no comprehensive curriculum coverage; no empirical data from South Asian contexts	Covers entire Grade 8 curriculum systematically; provides first empirical dataset from Sri Lankan classrooms through quasi-experimental study
SHILPA Project (Current Research) <i>B.A.S.Haath Bulathgama IT22913760 SLUIT</i>	Addresses all identified gaps: Curriculum alignment, low-resource optimization, South Asian validation, user-friendly design for struggling students	Comprehensive solution: Sri Lankan curriculum integration + offline-first architecture + empirical validation + accessible design for resource-constrained environments

Table 2: Research Gap

1.3 Research Problem

In Sri Lankan classrooms, mathematics is often taught through static, black-and-white textbook diagrams. A cube is shown with dotted lines to “suggest” depth, a parabola appears as a lifeless curve, and a bar chart remains frozen on the page. Students are expected to imagine the three-dimensional reality behind these sketches, but for many, this leap from flat drawings to abstract

mental models is too difficult. As a result, math learning often turns into rote memorization rather than true understanding.

This project asks how Augmented Reality (AR) can change that experience. It explores how an AR mobile app, built for low-end Android devices and able to run offline, can transform Grade 8 textbook diagrams into interactive 3D models and animations. The goal is to let students rotate, unfold, and experiment with shapes and graphs. At its core, the study seeks to discover whether giving students the power to “hold” mathematics in their hands can spark genuine curiosity and deeper learning.

<Finalized research question for the entire project>

<Identified/decomposed sub research questions (4) per each member>

<Proper highlight/indication of the particular members’ research question>

<NEW CHAPTER – NEW PAGE ALWAYS!!!!>

2 OBJECTIVES

2.1 Main Objective

To develop an Augmented Reality (AR) mobile application based on the Sri Lankan Grade 8 mathematics curriculum to enhance the learning experience of students by using interactive 3D simulations and overcoming the limitations of static textbook content.

2.2 Specific Objectives

Although international studies have shown again and again that Augmented Reality (AR) can improve how students learn, there is still a major missing piece: its use and testing in the Sri Lankan classroom. A closer look at the literature reveals four key gaps that this project is designed to **fill**. **Curriculum Alignment:** To investigate the Grade 8 Sri Lankan mathematics curriculum and identify challenging concepts in geometry (e.g., 3D objects, transformations), algebra (e.g., graphing functions), and data representation, and to map them to specific 2D diagrams in the government-issued textbook that will serve as AR markers.

1. **Interactive AR Application Development:** To create a Unity-based AR mobile application that can recognize these textbook diagrams and instantly render interactive 3D models and animations, allowing students to manipulate and explore abstract concepts in real time.
2. **Optimization for Low-Resource Contexts:** To engineer the application to be highly efficient and perform smoothly on low-end Android devices with 2-3GB of RAM, and to implement an offline-first architecture with preloaded assets to ensure full functionality without an internet connection.
3. **Empirical Evaluation:** To conduct a quasi-experimental study to measure the impact of the AR application on students' academic performance and conceptual understanding

in Grade 8 mathematics, using pre- and post-tests to generate the first localized dataset on AR's effectiveness in Sri Lanka.

4. **User Experience and Usability:** To design a highly user-friendly interface that simplifies navigation and interaction, making the application intuitive for Grade 8 students and their teachers, and to validate its usability through systematic testing.
5. **Knowledge Contribution:** To document and disseminate the findings of the study to fill a significant gap in the global literature on the application and effectiveness of AR in underrepresented South Asian educational contexts.

3 METHODOLOGY

3.1 Overall System Description

The system will be built on Unity 3D, leveraging its scene graph architecture and spatial mapping features to import high-fidelity 3D models and create immersive AR experiences. The app's core function is straightforward: a student will use a common smartphone or tablet to scan specific 2D diagrams directly from their physical Grade 8 mathematics textbook. This approach creates a deeply immersive and interactive learning environment where abstract concepts become tangible and easy to explore. It shifts learning from a passive act of observation to an active process of discovery.

The overall research will be structured around a Design-Develop-Evaluate (DDE) framework, comprising three distinct phases to ensure a systematic and evidence-based approach to project execution.

3.2 Technologies to be Used

T We picked our tech stack with care. It builds a strong, useful augmented reality app. Unity 3D acts as the main platform. We chose it for solid AR tools and good asset handling.

On the AR side, we can go with Unity's AR Foundation or Vuforia. AR Foundation is Unity's own setup, works across platforms. Vuforia stands out as a solid kit, especially for marker tracking that you can count on. That fits our plan perfectly. We're using textbook diagrams as those markers to bring up the digital stuff. This setup lets us pick whatever works best for what we need.

Blender handles the 3D models. It's pro-level software, free too. You get high-quality assets from it, optimized right. That's key for AR that runs smooth, even on weaker phones.

The app targets Android phones. We optimize hard for low-end ones. This choice ties right into the project's heart. We make it work on just 2-3 GB RAM. That way, students in Sri Lanka get ino matter if they're in cities or way out in rural spots with basic gear. The app becomes a tool for all, not just the lucky few



Figure 1



Figure 2

3.3 Research Design

The research follows a **Design–Develop–Evaluate** framework:

- **Design Phase:** Review the Grade 8 mathematics curriculum, identify abstract concepts, and design interactive 3D models.
- **Develop Phase:** Build the AR mobile app in Unity with marker-based tracking, preloaded assets, and offline features.
- **Evaluate Phase:** Conduct a quasi-experimental study in classrooms. Students will complete pre-tests and post-tests to measure improvement in comprehension, problem-solving, and confidence. Observations and feedback will provide qualitative insights.

3.4 Development & Testing Phases

- **Prototype Development:** Build a minimal version with sample geometry models.
- **Core Development:** Implement curriculum-specific 3D models, animations, and algebra tools.
- **Testing:**
 - *Functional Testing* – Ensure scanning, rendering, and interaction features work correctly.
 - *Usability Testing* – Test with a small group of students for clarity and ease of use.
 - *Performance Testing* – Check responsiveness on low-end Android devices.
 - **Classroom Pilot Study:** Deploy the app in real classrooms and compare results with traditional teaching.

3.5 Description of Personal and Facilities

As a final-year undergraduate student specializing in Interactive Media within the Information Technology program at SLIIT, I bring relevant coursework experience in 3D modeling, game development, and user interface design. My academic background includes completed projects in Unity development, AR application creation, and interactive system design, providing the foundational skills necessary for this research project.

Personal technical competencies include proficiency in Unity 3D development gained through academic coursework and personal projects, basic 3D modeling experience using Blender acquired through multimedia courses, and familiarity with AR development principles learned through specialized electives. Previous experience with C# programming and basic shader development provides the programming foundation necessary for custom AR interaction implementation.

The SLIIT Department of Information Technology (Interactive Media unit) provides comprehensive access to computer laboratories equipped with high-performance workstations suitable for 3D modeling and Unity development. These facilities include modern graphics cards capable of handling complex 3D scenes, sufficient RAM for large asset libraries, and high-resolution displays necessary for detailed design work and testing.

AR hardware requirements will be met through the university's existing equipment assets, including mobile devices and AR headsets available through the department. This arrangement eliminates the need for personal hardware investment while ensuring access to appropriate development and testing equipment throughout the project duration.

Software requirements leverage primarily free and student-licensed tools to minimize costs. Unity Personal Edition provides all necessary AR development features at no cost for academic projects. Blender offers comprehensive 3D modeling capabilities as open-source software, while other software is available through student licensing at minimal cost through academic programs.

4 PROJECT REQUIREMENTS

<Proper description – at least worth 200 words/2 paras needed)

4.1 Functional Requirements

- Ability to scan Grade 8 textbook diagrams.
- Marker-based AR tracking to display 3D models.
- Interactive features: rotate, unfold, slice 3D objects, adjust variables in graphs.
- Offline functionality with preloaded assets.
- Simple user interface suitable for school students.
- Basic feedback module to test and reinforce understanding.

4.2 Non-Functional Requirements

- **Performance:** Smooth operation on low-end Android devices (2–3GB RAM).
- **Scalability:** Expandable to additional subjects or higher grades.
- **Usability:** Easy for both students and teachers, minimal training required.
- **Reliability:** Functions without internet connectivity.
- **Affordability:** Low-cost and free for students to access.

5 PROJECT TIMELINE

Phase	Tasks	Duration
Requirement Analysis	Curriculum review, requirement gathering	2 weeks
Design	System design, UI/UX, prototype 3D models	4 weeks
Development	Unity AR setup, model creation, offline optimization	8 weeks
Testing	Functional, usability, performance	4 weeks
Classroom Pilot	Deployment in schools, pre/post evaluation	6 weeks
Data Analysis & Report	Analyze results, finalize documentation	4 weeks

Table 3 : Project Timeline

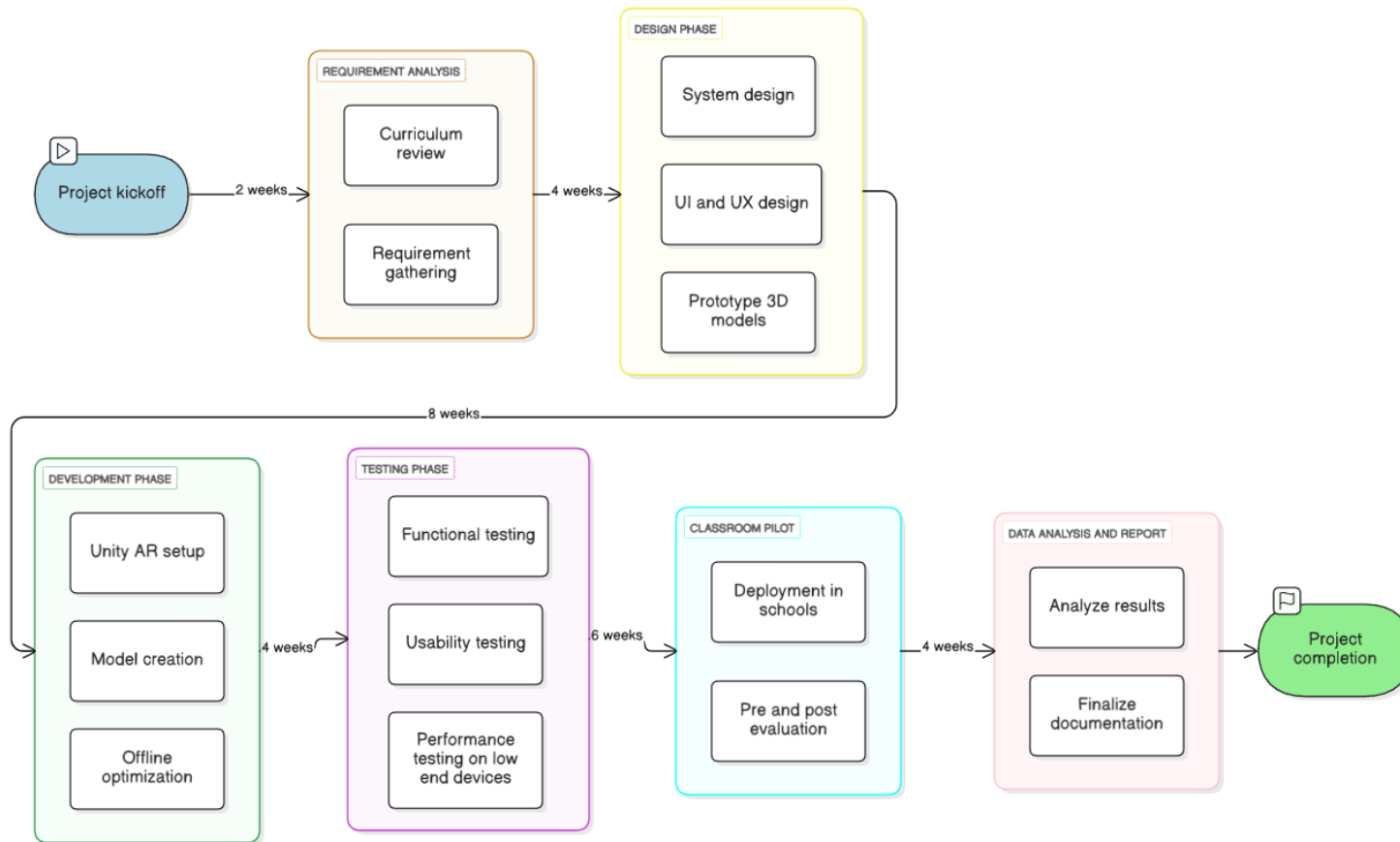


Figure 3: Work Breakdown Chart

Project Gantt Chart



Figure 4: Project Gantt Chart

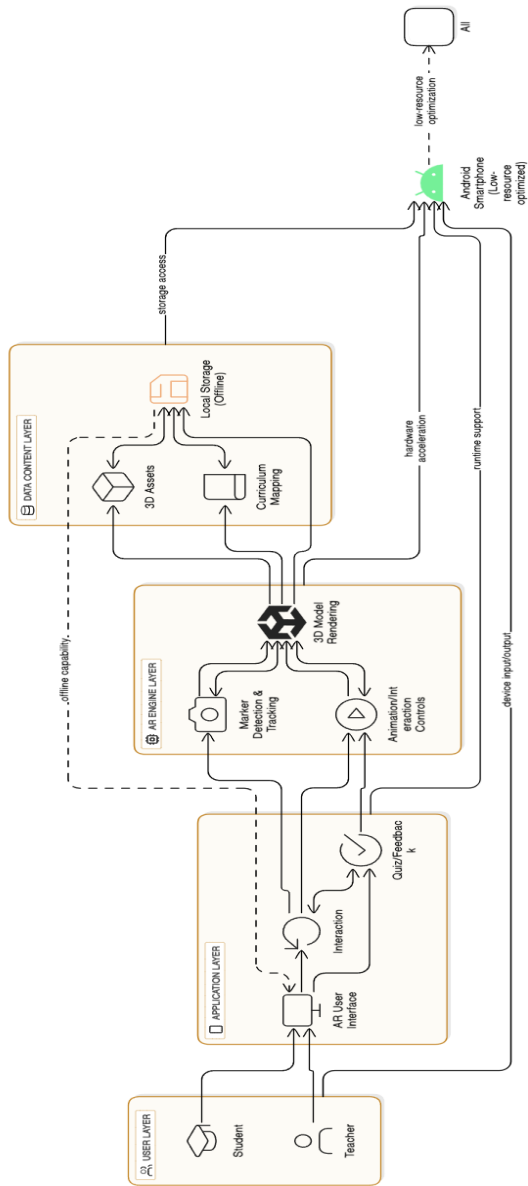


Figure 5: Single Component System Diagram

6 BUDGET AND BUDGET JUSTIFICATION

Investment Category	Cost Breakdown (LKR)	Pricing & Recovery Model (LKR)
Labor Costs	Development: LKR 2,500,000 Research: LKR 800,000 Testing: LKR 500,000	Freemium Model: Free: Basic Grade 8 content Premium: LKR 900/month full access
Material Costs	3D Assets: LKR 600,000 Software Licenses: LKR 300,000 Hardware/Testing: LKR 400,000	Institutional Licensing: School License: LKR 60,000/year District License: LKR 300,000/year
Marketing & Distribution	Digital Marketing: LKR 800,000 Partnership Development: LKR 500,000	Government Contracts: Ministry of Education: LKR 15,000,000/year Provincial Councils: LKR 4,500,000/year each

Table 4: Budget Justification

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